



OPEN Parkinson's disease severity clustering based on gait activity from mobile device

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Parkinson's disease (PD) is a neurodegenerative disorder characterized by motor symptoms, including gait impairments, which significantly affect patient mobility and quality of life. An accurate assessment of the severity of PD is crucial for clinical management. This study investigates the utility of smartphone-derived gait data to objectively cluster PD severity using unsupervised machine learning, with the aim of improving precision in disease monitoring. We analyze gait data from the mPower dataset, comprising 8779 accelerometer recordings from 1957 participants (PD patients and healthy controls). Stride cycles were segmented using frequency analysis and peak detection, followed by sequence padding to standardize input lengths. K-means clustering with dynamic time warping (DTW) was applied to identify gait patterns, while autoencoder embeddings and t-SNE visualized high-dimensional data. The groups were correlated with the MDS-UPDRS scores (Parts I and II) to assess severity. Four distinct gait clusters were identified, correlating with the severity of PD. The most severe group (Cluster 1) exhibited significantly higher MDS-UPDRS scores for balance/walking problems ($2.43\times$) and freezing episodes ($8.41\times$) compared to the least severe group (Cluster 4). The visualization of t-SNE confirmed the clear separation of the clusters, with higher severity scores concentrated in cluster 1. Sequence padding showed no significant impact on clustering outcomes ($p > 0.05$), validating its use for handling variable-length data. This study demonstrates that smartphone-derived gait patterns, analyzed via unsupervised clustering and visualization techniques, effectively stratify PD severity. Gait features related to balance, freezing, and walking difficulties are critical biomarkers for disease progression. Key advantage of our technique is the use of unsupervised learning to identify latent patterns without preconceived group assumptions, allowing subgroups to emerge organically and providing an unbiased exploration of gait-pattern relationships with Parkinson's severity. While limitations include the reliance on self-reported MDS-UPDRS data and k-means algorithm variability, these findings highlight the potential of wearable sensors and machine learning to develop objective, scalable tools for PD assessment.

Keywords Parkinson's disease, Severity clustering, Gait analysis, MPower dataset

Parkinson's disease (PD) is a neurodegenerative disorder that involves the gradual degeneration of neural tissues and primarily impacts the motor aspect of the body, characterized by tremors, rigidity, bradykinesia, and postural instability. In addition to the aforementioned motor manifestations, PD also affects a wide range of non-motor symptoms such as cognitive decline, affective disorders, autonomic nervous system disorders, and sleep problems, all of which add to the already overwhelming burden of the disease on patients' caregivers. Among the motor symptoms, abnormalities of gait are the most severe because of their critical effect on mobility, independence, and general life quality among patients suffering from PD^{1–3}. Gait impairments in PD are characterized by reduced speed when walking, shuffling steps, and freezing episodes as well as certain balance difficulties, which make them a priority of both clinical evaluation and treatment.

The progression and the severity of symptoms in PD differ amongst patients. Factors like age, onset, genetics, and the environment influence the condition, which makes it chronic and progressive. Clinicians rely on standardized rating scales to quantify PD symptoms, aiding in diagnosis, treatment planning, and research.

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One of the most widely used tools is the Unified Parkinson's Disease Rating Scale (UPDRS), which structured into four main sections to provide a comprehensive evaluation of various dimensions of PD. Part I evaluates non-motor experiences of daily living, covering aspects such as mood, cognition, and social interactions. Part II focuses on motor experiences of daily living, addressing the functional limitations encountered in routine activities. Part III involves a detailed motor examination, scoring specific motor signs like tremor, rigidity, and bradykinesia. Part IV assesses motor complications associated with long-term dopaminergic therapy, including fluctuations and dyskinesias. The Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) represents an updated version of the original UPDRS, designed to enhance its reliability and validity. While retaining the core structure of the UPDRS, the MDS-UPDRS introduces several improvements, such as refined item wording, expanded scoring options, and enhanced inter-rater reliability. Other scales that are commonly in use for measuring health-related quality of life are the PDQ-8 and the Hoehn and Yahr Scale (HY), which is a widely used clinical tool for staging the progression of PD based on the extent of functional disability and motor symptoms.

The use of wearable inertial measurement unit (IMU) sensors, which integrate accelerometers and gyroscopes, has gained significant attention in the diagnosis and monitoring of PD^{4–8}. These devices offer a non-invasive and cost-effective means to capture motion data, enabling the analysis of motor symptoms such as gait abnormalities, tremors, and bradykinesia. For instance,⁹ demonstrated the potential of frequency analysis applied to accelerometer data collected from wrist-worn sensors to classify PD patients. Trabassi et al. aimed to identify the most accurate supervised machine learning algorithm for classifying individuals with PD using a minimal set of IMU-derived gait features, with a sample size of 81 PD and 80 HS¹⁰. After extracting 22 gait features and applying a three-level feature selection process, seven key features were used to train five ML algorithms, including support vector machine (SVM), decision trees (DT), and random forest (RF). The results demonstrated that SVM, DT, and RF achieved the highest classification accuracy, exceeding 80% on the test set, while the proposed approach minimized overfitting and enhanced model explainability by addressing multicollinearity among gait features. Another study comprehensively analyzes the use of IMU-based gait analysis for assessing PD, utilizing data from 25 PD patients and 25 age-matched healthy individuals¹¹. The researchers evaluated 18 different configurations of gait parameters collected from eight IMU sensors, processed through six machine learning techniques, and incorporated two meta-classifiers based on majority voting with weighting schemes. Classification accuracy ranged from 63% to 80% across individual classifiers, with one meta-classifier configuration achieving up to 96% accuracy, highlighting the effectiveness of statistical preselection processes in optimizing performance. Notably, when reducing the number of sensors, features based on joint range of motion, particularly knee range of motion measured using four bilateral IMUs, outperformed spatio-temporal parameters, providing data-driven insights for selecting optimal sensor configurations and ML methods in PD severity grading.

In addition to classification methods, researchers have explored advanced analytical techniques to extract meaningful insights from sensor data¹². For instance,¹³ proposed a method to analyze stride cycles using accelerometer data, introducing the concept of biphasicity scores to quantify periodic patterns in gait. Similarly,¹⁴ developed an innovative approach leveraging self-supervised learning to model healthy gait patterns. Moreover, previous studies have applied autoencoder-based methods to gait signals, such as convolutional autoencoders trained on healthy activities of routine to extract transferable latent features for PD classification¹⁵, unsupervised denoising autoencoders for detecting freezing of gait episodes¹⁶, and volumetric autoencoders for anomaly detection in Parkinsonian gait video¹⁷. Their method encodes accelerometer and gyroscope data into a latent space, creating a representation of normal walking behavior. By applying anomaly detection algorithms to this latent space, they successfully identified deviations indicative of PD, showcasing the potential of unsupervised learning in detecting pathological gait patterns.

However, one recurring limitation in such studies is the small sample size of volunteers, a common challenge in medical research that can hinder the generalizability of findings. To address this issue, crowdsourced datasets like mPower have emerged as valuable resources. The mPower dataset contains extensive accelerometer and gyroscope data collected from a large population of participants, including both healthy individuals and PD patients. This wealth of data helps mitigate the problem of small sample sizes and provides a robust foundation for developing and validating machine learning models. For example,¹⁸ and¹⁹ utilized the mPower dataset to classify PD patients by extracting and preprocessing features from accelerometer and gyroscope data. Previous works employed various machine learning techniques to achieve accurate classification results, demonstrating the potential of data-driven approaches in PD research²⁰.

Building on these advancements, our work aims to investigate the relationship between gait cycles and MDS-UPDRS scores using the mPower dataset. Specifically, we concentrate on the clustering of cycles of gait from walking tests, which have participants walking in a straight line and u-turn while their smartphones are in their pockets. This setup allows us to capture naturalistic gait patterns in a real-world setting. By developing and implementing innovative approaches to cluster and extract features from gait cycles, we hope to identify features that are associated with the clinical measurements of the severity of PD as determined by the MDS-UPDRS scores. Our ultimate goal is to contribute to the development of more precise and interpretable tools for assessing PD progression and severity, which will lead to better care and patient outcomes.

Material and methods

Database and acquisition setup

The mPower: Mobile Parkinson Disease Study dataset²¹ represents one of the largest mobile application-based studies dedicated to PD research. Designed as a tool for monitoring key indicators of PD progression, the mPower app leverages smartphone technology to collect data from a large, longitudinal cohort of participants, including both individuals with PD and healthy controls. This approach has enabled the recruitment of a diverse and

geographically dispersed population, facilitating extensive data collection over time. The dataset encompasses seven distinct modules: Demographics, MDS-UPDRS, PDQ-8, Memory, Tapping, Voice, and Walking. In this study, we specifically focus on the Walking and MDS-UPDRS modules, which provide critical insights into motor function and disease severity.

The walking module leverages the IMU sensor embedded in a mobile phone to collect gait data. The IMU sensor typically comprises an accelerometer, gyroscope, and pedometer, which work together to capture detailed information about movement patterns, such as acceleration, angular velocity, and step counts. These metrics provide valuable insights into gait characteristics, enabling the analysis of motor function in individuals with PD. In parallel, the severity of PD symptoms is assessed through the MDS-UPDRS module. The MDS-UPDRS questionnaire is structured into four distinct parts: Part I evaluates non-motor experiences of daily living, Part II focuses on motor experiences of daily living, Part III involves a clinician-scored motor examination, and Part IV assesses motor complications [[6]]. However, in the dataset utilized for this study, the majority of the available data pertains only to Parts I and II, which are specifically designed to capture the patient's experiences in daily living, both non-motor and motor. Given this limitation, our analysis focuses exclusively on these two parts, as they provide critical insights into the functional impact of PD on routine activities. By combining gait data from the walking module with symptom severity scores derived from Parts I and II of the MDS-UPDRS, we aim to establish meaningful correlations between objective gait metrics and subjective clinical assessments.

The mPower dataset includes 35,410 walking test samples from 3,101 unique participants and 2,305 MDS-UPDRS survey samples from 2,024 unique participants. The disparity in sample sizes between the MDS-UPDRS survey and walking data can be attributed to the survey's design, which requires participants to complete it only once a month. In contrast, walking tests can be performed more frequently. To analyze the relationship between gait patterns and disease severity scores, we integrated these two datasets by matching participants' unique IDs and aligning MDS-UPDRS scores with the walking activity dates closest to the survey completion. To ensure data quality, we estimated the number of steps using stride extraction techniques and cross-verified these estimates with pedometer data. Samples with estimated step counts deviating by more than 20% from the pedometer data were excluded as potentially inaccurate. After preprocessing, a total of 8,779 high-quality samples from 1,957 unique participants were retained for clustering and visualization analysis. The dataset included both PD patients and healthy controls, with a PD-to-control ratio of approximately 1:2. Across all participants, the average age was 43.4 ± 18.3 years, 70.2% were male, and 34.2% reported smoking history. The mean MDS-UPDRS scores were 4.9 ± 3.8 for Part I and 3.4 ± 4.9 for Part II. Other clinical measures, such as Hoehn & Yahr staging, were not available in the dataset.

The mPower dataset provides accelerometer and gyroscope signals, which we utilized to segment individual strides from the accelerometer data. Since the dataset explicitly separates walking tasks into straight walking and combined resting/turning phases, only the segments labeled as straight walking were included in this study. Over the years, several gait segmentation techniques have been proposed, including zero-crossing methods, multi-subsequence dynamic time warping (DTW), hierarchical hidden Markov models (HMM), and peak detection approaches. However, techniques like multi-subsequence DTW^{22–25}, and HMM^{26–28} often require pre-labeled stride events to serve as subsequence templates during the learning process, which can be labor-intensive and limit their scalability. In contrast, peak detection methods offer a more flexible alternative, with threshold-based approaches being one of the most widely used strategies for identifying peaks and segmenting stride events. In this study, we employed frequency analysis to accurately determine the location of peaks, enabling robust and automated stride segmentation without the need for manual labeling.

Figure 1 illustrates an overview of our system. Raw accelerometer signals are first processed through stride extraction to identify individual strides. Next, the extracted sequences undergo sequence padding to standardize their lengths, ensuring consistency and uniformity for subsequent analysis. Both padded and unpadded sequences are then clustered using k-means clustering, with DTW serving as the distance metric. We noted that K-means clustering represents the TimeSeriesKMeans from tslearn library. This approach allows us to group similar stride patterns effectively. To assess the impact of padding on the results, we conducted a comparative analysis between padded and unpadded sequences. We then performed statistical analysis on the clustered data and employed visualization techniques such as autoencoder embedding and t-SNE dimensionality reduction. These methods transform the complex, high-dimensional data into a more manageable format, enabling us to visualize and understand the relationships between different gait patterns and PD severity.

The PDKit²⁹, a toolkit for the digital assessment of PD, was used in this study to process accelerometer data via frequency analysis and provide subsequence segmentation at the step level. Since our objective was stride-level segmentation for clustering, we combined two consecutive steps into a single stride cycle. To ensure proper pairing of steps, we first applied k-means clustering ($k = 2$) with DTW as the distance metric, which allowed us to group steps according to temporal similarity. Valid strides were defined only as pairs of consecutive steps from different clusters, under the assumption that alternating clusters correspond to left and right steps. Consecutive steps from the same cluster were excluded, as they likely represent the same-foot steps or misclassifications. Finally, to maintain data quality, we validated stride counts against pedometer readings and removed samples where estimated step counts deviated by more than 20%, as previously described.

Sequence padding

For visualization purposes, we transformed the stride subsequences into a latent space using an autoencoder model. However, a common limitation of simple autoencoders is their requirement for input data to have a consistent range, which poses a challenge given that our stride candidates exhibit varying lengths. To address this issue, we implemented the sequence padding algorithm outlined in Algorithm 1, which adjusts the sequence length to ensure uniformity. In this study, the padding length was set to the third quartile of the stride candidates' lengths, as this approach minimizes distortion while accommodating the majority of the data. The impact of this

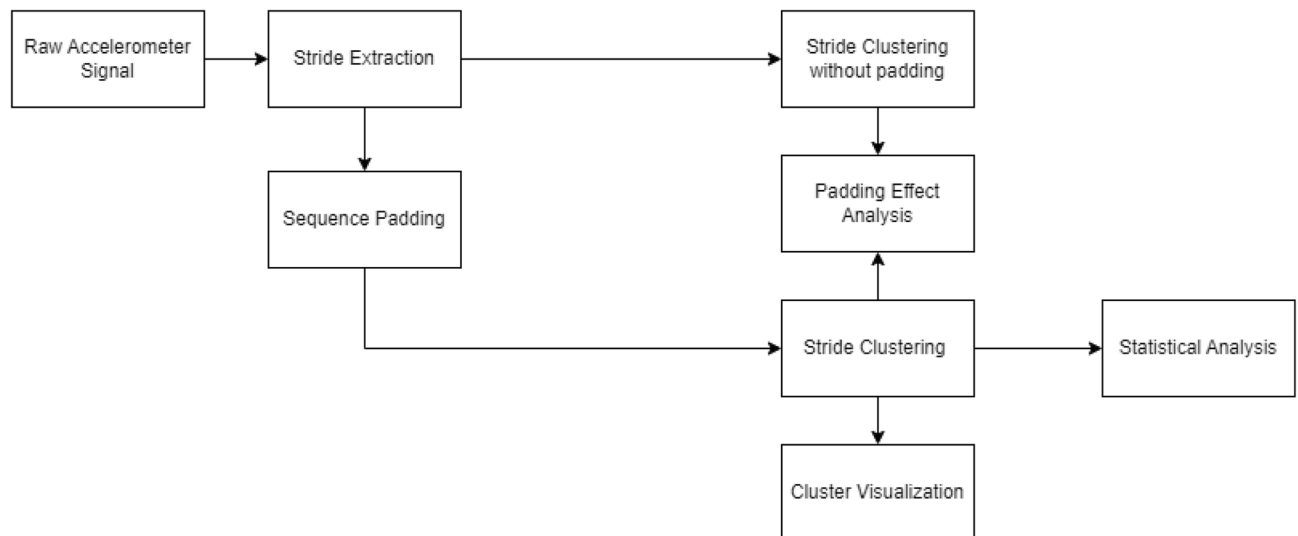


Fig. 1. Overview of the system workflow, illustrating the processing pipeline from raw accelerometer signals to cluster visualization.

sequence padding on the quality of the latent space representation will be further analyzed and discussed in the results section.

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Function SequencePadding (padding_length)
  Data: stride_subsequence
  Result: padded_sequence
  begin
    stride_subsequence_length = len(stride_subsequence)
    if stride_subsequence_length > padding_length then
      | remove the data which over the padding_length
    else
      | add 0 to the end of stride_subsequence until length of stride_subsequence equal to padding_length
  end
  
```

Algorithm 1. Sequence Padding

Stride clustering

In this work, we applied k-means clustering, an unsupervised learning algorithm designed to partition instances into k clusters by assigning each instance to the cluster with the nearest mean based on a distance metric. Similar method was used to successfully clustering PD severity based on tapping activity on mobile device³⁰. Unsupervised learning enables the exploration of latent patterns in the data without relying on preconceived assumptions about groupings. K-means clustering can identify distinct gait patterns that naturally emerge from the data, potentially uncovering subgroups of patients with similar gait characteristics but differing levels of disease severity. Moreover, while the mPower dataset includes MDS-UPDRS scores, these are self-reported and may introduce noise or bias, which could undermine the performance of supervised learning approaches that depend explicitly on severity scores. In contrast, unsupervised learning avoids reliance on potentially noisy or incomplete labels, allowing it to identify natural groupings in gait patterns more effectively. This makes it particularly well-suited for handling the complexity and variability of real-world gait data collected via smartphones.

The objective function, shown in Equation 1, minimizes the overall distances between each member and the centroid of its respective cluster. Here, x represents the stride data within cluster S_i , μ_i denotes the centroid of cluster i , and DTW serves as the distance function. DTW is a robust algorithm for measuring similarity between two temporal sequences, particularly advantageous when the sequences vary in speed or timing. In our study, we used k-means clustering with DTW to group stride candidates derived from accelerometer signals, aiming to identify patterns associated with disease severity. Based on the MDS-UPDRS survey scores, the number of clusters k was set to 4, corresponding to the four levels of severity. We tested different values of k ranging from 2 to 9. When k was set to fewer than 4, the resulting clusters failed to clearly separate distinct gait patterns associated with different severity levels. In contrast, increasing k beyond 4 produced overly fragmented clusters that were more difficult to interpret clinically. Thus, selecting $k = 4$ provided an optimal balance between resolution and interpretability for stratifying disease severity.

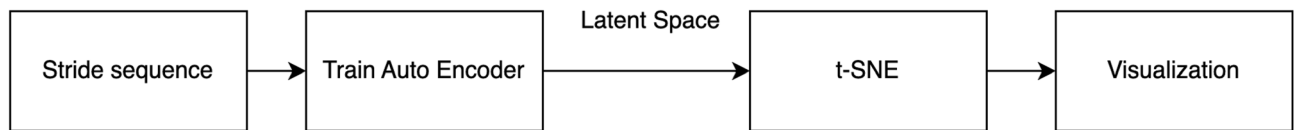


Fig. 2. Cluster visualization process using autoencoder model and t-SNE dimensionality reduction to transform high-dimensional gait data into a two-dimensional space for better interpretability.

Cluster	Severity score				
	MDS-UPDRS 2.10	MDS-UPDRS 2.12	MDS-UPDRS 2.13	Part 1	Part 2
1	1.05 ± 0.95	0.62 ± 0.73	0.24 ± 0.61	0.81 ± 0.54	0.64 ± 0.59
2	0.90 ± 0.85	0.37 ± 0.57	0.12 ± 0.45	0.68 ± 0.52	0.43 ± 0.48
3	0.80 ± 0.85	0.25 ± 0.49	0.06 ± 0.28	0.61 ± 0.46	0.39 ± 0.42
4	0.79 ± 0.69	0.25 ± 0.47	0.03 ± 0.16	0.59 ± 0.42	0.39 ± 0.38

Table 1. Average severity scores calculated for each cluster based on MDS-UPDRS questions; MDS-UPDRS 2.10 (tremor), MDS-UPDRS 2.12 (balance and walking problems), MDS-UPDRS 2.13 (freezing episodes), Part 1 (non-motor aspects of daily living, and Part 2 (motor aspects of daily living).

$$\arg \min_S \sum_{i=1}^k \sum_{x \in S_i} DTW(x, \mu_i) \quad (1)$$

Given that the primary objective of this research is to classify patient severity using gait accelerometer data, k-means clustering provides a suitable solution to achieve this goal. The algorithm groups stride cycles with similar characteristics into clusters, where each cluster represents a distinct pattern of gait behavior. By calculating the mean severity score of the MDS-UPDRS ratings within each cluster, we can infer the severity level associated with the gait characteristics of that cluster. This approach allows us to link objective gait metrics derived from accelerometer data to subjective clinical assessments of disease severity, thereby facilitating a data-driven classification of patient severity based on gait patterns.

Cluster visualization

To enhance the clarity and interpretability of our results, we incorporated a visualization process for the time-series gait data. Figure 2 illustrates our cluster visualization process. Following the approach outlined in previous work³¹, we utilized an autoencoder model to embed the time-series stride data into a latent space. The encoder consisted of three Dense layer with Relu activation function for each model. The loss is set to Mean Absolute Error and the optimizer is Adam. This latent representation captures the essential features of the data in a more compact form.

To make the clusters visually interpretable, we further applied manifold learning techniques to reduce the dimensionality of the latent space through t-SNE, enabling us to plot and analyze the clustering results in a two-dimensional space. t-SNE was particularly useful for capturing local structures in the latent space, making the cluster boundaries more distinguishable. Since the autoencoder model requires a fixed input size, as stated earlier, we addressed this constraint by padding the sequences to a uniform length before performing k-means clustering. This step ensures that all stride cycles, regardless of their original lengths, can be processed consistently by the model. By combining sequence padding, autoencoder embedding, and dimensionality reduction, we were able to effectively visualize the clustering results, providing deeper insights into how gait patterns relate to disease severity.

Results and discussion

Clustering severity score

For this experiment, we calculated the severity score for each cluster based on the MDS-UPDRS scores. The severity score of a cluster was determined by taking the mean of the MDS-UPDRS scores of its members. As shown in Table 1, each cluster was characterized using mean MDS-UPDRS scores across relevant items (2.10 tremor, 2.12 balance/walking problems, 2.13 freezing, Part I non-motor, and Part II motor aspects). Cluster 1 consistently exhibited the highest severity, while Cluster 4 showed the lowest, with significant differences observed particularly in walking, balance, and freezing scores. These results demonstrate that the unsupervised clusters correspond to clinically meaningful gradations of disease severity as captured by MDS-UPDRS. These pronounced differences are both statistically significant and pathophysiologically relevant. Freezing of gait and balance deficits are closely linked to malfunction in the fronto-basal ganglia circuits and anomalies in the dopaminergic pathways in PD, which are fundamental to motor initiation and postural regulation^{3,8}. The apparent distinction between severe and mild gait clusters thus reflects underlying neurodegenerative processes, implying that stride-level gait traits can function as early digital biomarkers of disease progression¹³.

Our findings are in agreement with previous studies that have linked freezing and balance impairments with PD severity, and further demonstrate that these clinically relevant distinctions can be identified through an unsupervised clustering framework applied at the stride level.

Considering the components of the MDS-UPDRS, specifically Part 1 (Non-motor aspects of experiences of daily living) and Part 2 (Motor aspects of experiences of daily living), our results suggest that gait-related features can effectively distinguish between the most and least severe groups. For instance, MDS-UPDRS item 2.10, which assesses tremor, revealed that the average score in the most severe cluster (Cluster 1) was 1.32 times higher than in the least severe cluster (Cluster 4). Similarly, MDS-UPDRS item 2.12, which evaluates problems with balance and walking, showed that the score in Cluster 1 was 2.43 times higher compared to Cluster 4. Most strikingly, MDS-UPDRS item 2.13, which focuses on freezing episodes or sudden inability to move when the feet feel “stuck” to the floor, demonstrated an even greater disparity, with the score in Cluster 1 being 8.41 times higher than in Cluster 4.

These findings underscore the significance of gait-related features, particularly those associated with balance, walking, and freezing episodes, as critical indicators of PD severity. Collectively, the evidence highlights the potential of gait analysis to provide valuable insights into the progression and functional impact of PD, reinforcing its importance in both clinical assessment and research settings.

Cluster visualization

Similar to Jiang et al.¹⁴, who used autoencoder-based representations to identify healthy from pathological gait, our latent space revealed separable subgroups corresponding to PD severity. Noor et al. found that denoising autoencoders could detect freezing events, which supports our observation that clusters with high freezing scores develop spontaneously from latent gait variables¹⁶. To gain a comprehensive understanding of the distribution of subjects in this experiment, we employed an autoencoder technique to explore the latent space representation of the clusters. Using t-SNE to reduce the dimensionality of the latent space into a two-dimensional scatter plot, we confirmed that the resulting clusters are visually distinguishable from one another, as illustrated in Fig. 3. In this plot, the x- and y-axes represent the mapped latent space dimensions. Figure 3a highlights the overall distribution of the clusters, clearly showing that Cluster 1 (red) is well-separated from Cluster 4 (green). Figure 3b provides further insight by visualizing the spread of individual sample points within each cluster, with color indicating the severity score. Notably, the group of samples with higher severity scores is concentrated in Cluster 1, which corresponds to the most severe group. This observation strongly supports the ability of the selected features to effectively separate subjects based on their disease severity, demonstrating the discriminative power of our approach.

To further demonstrate the clinical relevance of the identified clusters, Fig. 4 presents representative stride profiles derived from the average normalized waveforms of each cluster. The curves reveal distinct differences in gait patterns across severity levels. Notably, Cluster 1 (red), representing the most severe group, exhibits reduced amplitude in the early phase of the stride cycle, aligning with bradykinesia and diminished ground clearance frequently observed in patients with advanced Parkinson's disease. In contrast, Cluster 4 (green), the least severe group, displays higher and more consistent amplitudes, indicative of more stable and preserved gait patterns.

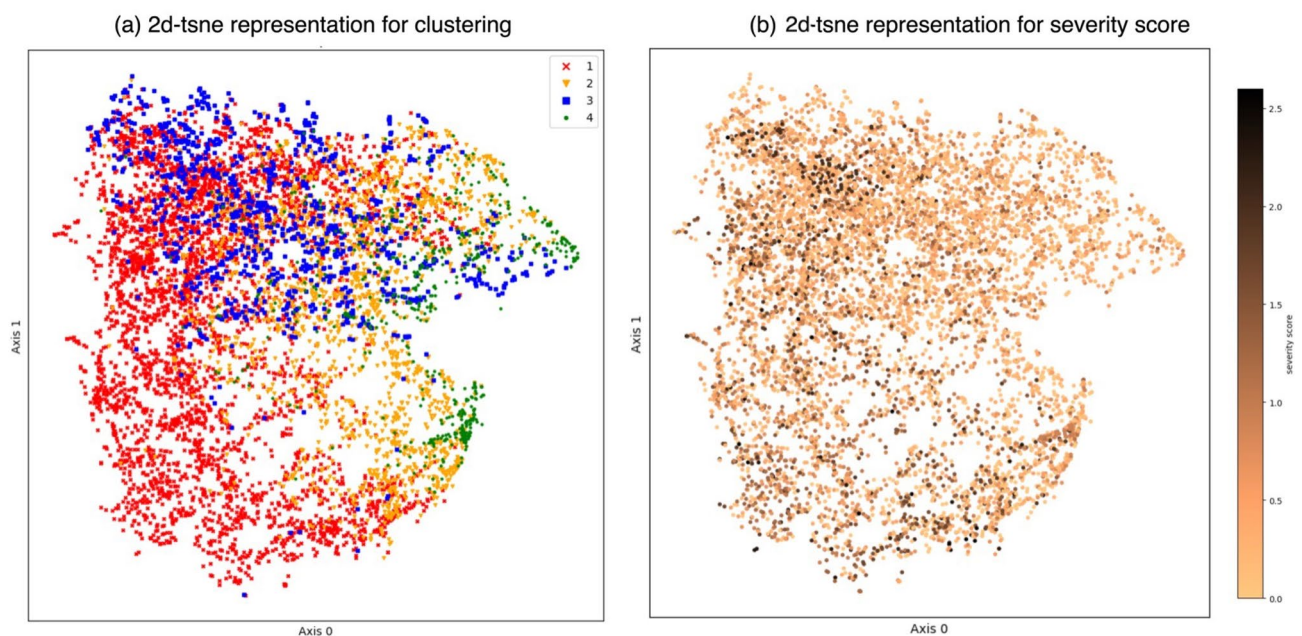


Fig. 3. Clusters latent space representation using t-SNE dimensional reduction. **(a)** Illustrates the overall distribution of clusters. **(b)** Provides individual sample points within each cluster, color-coded by severity score.

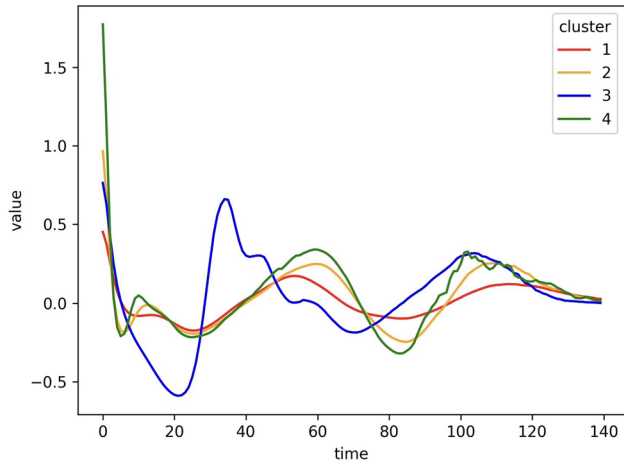


Fig. 4. Mean stride waveforms for each cluster. Cluster 1 (red), the most severe, shows reduced amplitude in the early stride phase, consistent with bradykinesia and diminished ground clearance, while Cluster 4 (green) maintains higher and more regular amplitudes, reflecting preserved gait function.

Cluster	p-value						Sample Size
	MDS-UPDRS 2.10	MDS-UPDRS 2.11	MDS-UPDRS 2.12	Part 1	Part 2	Total	
1	0.73	0.51	0.50	0.58	0.35	0.53	4465
2	0.58	0.55	0.99	0.88	0.96	0.61	1945
3	0.98	0.82	0.07	0.03	0.99	0.59	1031
4	0.84	0.92	0.01	0.18	0.89	0.68	561

Table 2. Hypothesis testing comparing the impact of padding versus non-padding sequences on the severity scores. The table lists p-values for different segments, indicating no significant difference between the two approaches.

Padding effect

Table 2 demonstrates that there is no significant difference between the use of padding and non-padding sequences in our analysis. Given that the padding sequence technique was applied to standardize input lengths in our study, it was crucial to validate whether this preprocessing step influenced the severity scores of the clusters. To assess this, we performed a t-test to evaluate the null hypothesis that there is no difference in the mean severity scores of the clusters when comparing padding and non-padding sequences. The results showed that the p-values for all segments were not statistically significant, indicating that the padding sequence technique does not have a meaningful impact on the severity scores. This finding confirms that the padding process preserves the integrity of the data and does not introduce bias into the clustering outcomes, thereby validating its use in our analysis. Compared to previous supervised and semi-supervised approaches, such as Nair et al., who utilized DTW-derived features for early Parkinson’s disease detection, and Meral et al., who achieved near-perfect staging with CNN-LSTM models, our unsupervised clustering framework provides distinct advantages^{23,32}. By not relying on pre-labeled severity groups, our method uncovers latent substructures in gait patterns that align coherently with clinical severity. This data-driven discovery empowers the identification of clinically meaningful subgroups that may remain undetected in traditional supervised pipelines, highlighting the strength of unsupervised methods for revealing novel and unbiased insights into gait-related biomarkers of PD.

Limitations

While this study provides valuable insights into the relationship between gait patterns and PD severity, it is important to acknowledge its limitations. First, the inherent randomness in the K-means clustering algorithm poses a potential limitation. Due to the random initialization process, the clustering results may not always converge to the global optimum 1. To mitigate this issue, we ran the algorithm 1,000 times and used two evaluation metrics to select the most robust clustering solution. However, further exploration of alternative clustering algorithms, such as hierarchical clustering or density-based methods, could provide additional validation and potentially improve the results. Second, the generalizability of our findings is limited by the specific characteristics of the mPower dataset. While the dataset offers a large and diverse sample size, the participants are self-selected volunteers who may not fully represent the broader PD population. Additionally, differences in data collection methods, smartphone models, and user compliance could introduce variability that affects the reliability of the results. Future studies

should validate these methods using other datasets or cohorts with standardized data collection protocols to ensure broader applicability.

By addressing these limitations in future work, researchers can further refine the methodologies and broaden the applicability of this study. Importantly, while our clustering results are comparable to previous studies that used supervised classification or autoencoder-based anomaly detection^{10,14}, our unsupervised approach provides a unique perspective by revealing latent subgroups that lack predefined severity labels. Unlike previous research, which depended on labeled data, our strategy emphasizes data-driven discovery. Future work can directly compare these approaches to establish the optimal strategies for clinical translation.

Conclusion

In this study, we investigated the relationship between gait patterns and PD severity using data from the mPower dataset. By applying k-means clustering with DTW, we identified distinct subgroups of gait patterns that correlate with varying levels of PD severity as assessed by MDS-UPDRS scores. One key advantage of our technique is that unsupervised learning enables the identification of latent patterns within the data without relying on preconceived assumptions about group structures or classifications. This approach allows for the organic emergence of distinct subgroups based on inherent data characteristics, offering a more unbiased and exploratory perspective on the relationships between gait patterns and PD severity. The combination of autoencoder embeddings and t-SNE dimensionality reduction facilitated the effective visualization of high-dimensional gait data, allowing for greater insight into the interrelationships between gait features and PD severity. The representative stride waveforms reinforce the clinical validity of the clusters, as the reduced amplitude observed in the most severe group (Cluster 1) is consistent with bradykinesia and reduced ground clearance typically seen in advanced Parkinson's disease. The use of sequence padding to standardize input lengths for clustering and visualization was validated, showing no significant impact on the severity scores, thus confirming its suitability for handling variable-length stride data.

The clinical implications of this study are encouraging, suggesting potential to enhance the assessment and management of PD. The demonstrated ability to stratify PD severity based on smartphone-derived gait data provides clinicians with a powerful, scalable tool for monitoring disease progression remotely and continuously, overcoming the limitations of traditional episodic clinical assessments. Clinically, the distinct gait clusters could facilitate personalised patient management by identifying individuals who are likely to benefit from earlier or intensified interventions, such as specific physiotherapy or tailored pharmacological strategies to mitigate balance problems and freezing episodes. Moreover, this method holds promise for enriching patient stratification in clinical trials, potentially improving trial efficiency by enabling more precise patient selection and outcome measurement based on objectively defined severity clusters. Ultimately, integrating gait clusters into decision-support platforms could significantly enhance personalised care, allowing clinicians to pre-emptively address symptoms, minimise complications such as falls, and optimise therapeutic interventions, thereby improving patient quality of life and reducing healthcare utilisation.

Despite the promising results, our study acknowledges certain limitations. The inherent randomness of the k-means algorithm and the self-reported nature of the mPower dataset may affect the generalizability of our findings. To further validate the findings of this study, future research should employ controlled, prospective data collection methods that ensure standardized protocols for participant recruitment, data acquisition, and analysis. Additionally, integrating complementary digital biomarkers such as voice or hand motion could provide a more comprehensive assessment of PD severity. For instance, our prior work demonstrated the utility of tapping activity on mobile devices for clustering PD severity³⁰. Combining gait analysis with other modalities may enhance the precision and robustness of severity stratification, offering a multimodal framework for monitoring disease progression. These strategies would not only enhance the reliability of the data but also pave the way for broader adoption of digital biomarkers in real-world clinical settings.

Data Availability

These data were contributed by users of the Parkinson mPower mobile application as part of the mPower study developed by Sage Bionetworks and described in Synapse [<https://www.synapse.org/mPower>]. The preprocessed datasets generated for this study are available on request to the corresponding author.

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Author contributions

DS designed the experiments and was involved in all facets of the study and manuscript preparation. PS and PA performed the experiments, data analysis, and manuscript preparation. WA was responsible for data extraction and processing for analysis. RB was involved in manuscript preparation. All authors approved the final manuscript.

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Declarations

Ethics Statement

The studies involving human participants were reviewed and approved by the local Ethics Committees of the Western Institutional Review Board as stated by Bot et al. (2016). Informed consent was obtained from all the participant. Ethical oversight of the study was obtained from Western Institutional Review Board and all methods were carried out in accordance with their guidelines and regulations. Before completing the e-consent process, prospective participants had to pass a five-question quiz evaluating their understanding of the study aims, participant rights, and data sharing options. Their signed consent form was sent to the provided email addresses for verification of their enrollment in the study. There is no additional ethical approval independently for this study. The data contributors allows qualified researchers to analyse the data and publish

the findings in open access publications.

Competing interests

The authors declare no competing interests.

Additional information

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